

Labwork 3 Report

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1 Implementation & Results

For this assignment, we use An RGB image of size 4536×8064 to load and reshape it into $36,578,304 \times 3$ RGB values. Grayscale conversion was implemented using a CPU for-loop and a Numba CUDA kernel. The `time.time()` function was used to execute time measurement.

The CPU execution time was approximately 34.71 s and the GPU execution time was approximately 0.05 s, the execution time of GPU is faster than CPU $694.2\times$.

2 Block Size Experiment

We also ran a series of tests to see how tweaking the CUDA block size affects overall kernel execution. By adjusting the thread block size from 8 to 512, we see a clear trend in which the execution time drops noticeably at the start—falling from 0.075 seconds to around 0.0491 seconds.

The sweet spot in our setup was a block size of 512, which recorded the fastest execution time at 0.0491 seconds. Once the block size reached 64 and above, the performance gains flattened out and the execution times remained fairly steady.

